secondary ionization is often larger than that of the primary ionization but the average kinetic energies of the secondary electrons are lower than those of the primary electrons. In many interactions the initial radiation transfers insufficient energy for ionizations; instead an electron is raised to a higher, excited energy level of the atom. These *excited atoms* rapidly return to lower energy states by emission of electromagnetic radiation such as X-rays, visible light, etc. For neutrons the absorption process involving the capture of the neutron imparts sufficient recoil energy to cause ionization and excitation.

7.2. ABSORPTION CURVES

In order to measure the absorption of nuclear radiation, the experiments must be performed in such a manner as to eliminate as many of the interfering factors as possible. Usually a well-collimated beam is used. This is illustrated in Figure 7.1 for a point radioactive source. The relation between the disintegration rate A and the count rate R is given by Eqn (7.1):

$$R = \psi A \tag{7.1}$$

The *counting efficiency* ψ includes a number of factors:

$$\psi = \psi_{\text{sample}}\, \psi_{\text{abs}}\, \psi_{\text{det}}\, \psi_{\text{geom}} \tag{7.2}$$

If conditions were ideal, there would be no self-absorption or scattering in the sample (in which case $\psi_{\text{sample}} = 1$), no absorption of radiation between the sample and the detector window ($\psi_{\text{abs}} = 1$), and the detector would have a 100% efficiency (sensitivity) to a "count" for each particle reaching its window ($\psi_{\text{det}} = 1$).

The geometric efficiency ψ_{geom}, is 1 for 4π-geometry, i.e. for a spherical detector subtending a 360° solid angle about the sample. Although such detectors exist (Ch. 9), more commonly the sample is counted outside the detector at some distance r, as indicated in Figure 7.1. If the detector window offers an area of S_{det} perpendicular to the radiation, the geometrical efficiency is approximated by (for small ψ_{geom})

$$\psi_{\text{geom}} \approx S_{\text{det}} / \left(4\pi r^2\right) \tag{7.3}$$

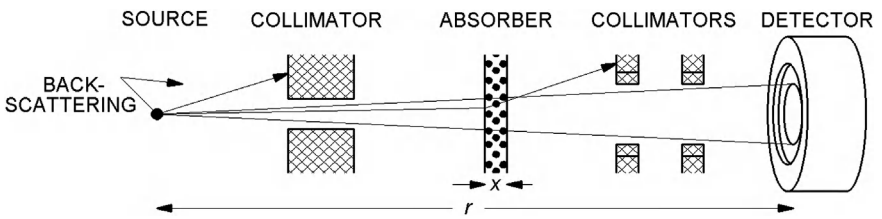


Figure 7.1 Geometrical arrangement for measuring absorption curves.

If a detector with a circular window of radius s is at a distance r from a source of radius d, the geometrical efficiency is given by

$$\psi_{\text{geom}} = \tfrac{1}{2}\left[1 - \left(1 + s^2/r^2\right)^{-\frac{1}{2}}\right]k \tag{7.4}$$

When the sample is a point source, $k = 1$, else k can be read from the series of curves in Figure 7.2.

The activity measured is proportional to the particle flux φ reaching the detector

$$R = k_{\text{det}}\,\varphi \tag{7.5}$$

where $k_{\text{det}} = S_{\text{det}}\,\psi_{\text{det}}$ and

$$\varphi = \psi_{\text{abs}}\,\varphi_0 \tag{7.6}$$

and

$$\varphi_0 = \psi_{\text{sample}}\,nA/\left(4\pi r^2\right) \tag{7.7}$$

where φ_0 is the flux of particles (particles $\text{m}^{-2}\,\text{s}^{-1}$) which reach the detector from the source when $\psi_{\text{abs}} = 1$, and n is the number of particles emitted per decay ($n > 1$ only for γ following α- and β-decays). Thus if every β-decay yields 2γ (in cascade) and γ is the measured radiation, then $n_\gamma = 2$, and $\varphi_0 = n_\gamma A_\beta/(4\pi r^2)$ γ-quanta $\text{m}^{-2}\,\text{s}^{-1}$. In branched decay n will not be an integer. Equation (7.7) is the so called $1/r^2$ - *law* since the measured flux varies as the inverse of the square of the distance to the source.

These equations are valid as long as the conditions at the source and at the detector, as well as r, are kept constant. When an absorber is inserted between the source and detector (Fig. 7.1), ψ_{abs} depends on the absorber thickness x (m). For zero absorber thickness, $\psi_{\text{abs}} = 1$ in accurate measurements. There is a small absorption due to the air between the sample and detector unless the measurement is done in a vacuum.

Absorption curves relate the variation of either R or φ to the thickness of the absorbing material. In Figure 7.3 the relative transmission φ/φ_0 is plotted as a function of absorber thickness for different kinds of radiation. For charged particles, i.e. electrons, protons, and heavier ions, φ/φ_0 reaches zero at a certain x-value (x_{max}); this is referred to as the maximum range of the particles. The range can be expressed by either the *average range* ($x = C_1$ for heavy ions and C_3 for electrons) or the *maximum* (or extrapolated) *range* (C_2 and C_4, respectively, in Fig. 7.3). The loss of energy involves collisions with atomic electrons, and the energy loss per collision and the number or collisions varies from one ionizing particle to the next, resulting in a slight *straggling* in the range. The average range is the meaningful one.

Figure 7.4 shows an absorption curve for radiation from ^{32}P. The radioactivity R has been measured as a function of aluminum absorber thickness in *linear density* ($\text{kg}\,\text{m}^{-2}$ in SI-units, sometimes also in $\text{mg}\,\text{cm}^{-2}$). The low activity "tail" (C_4 in Fig. 7.4) is the

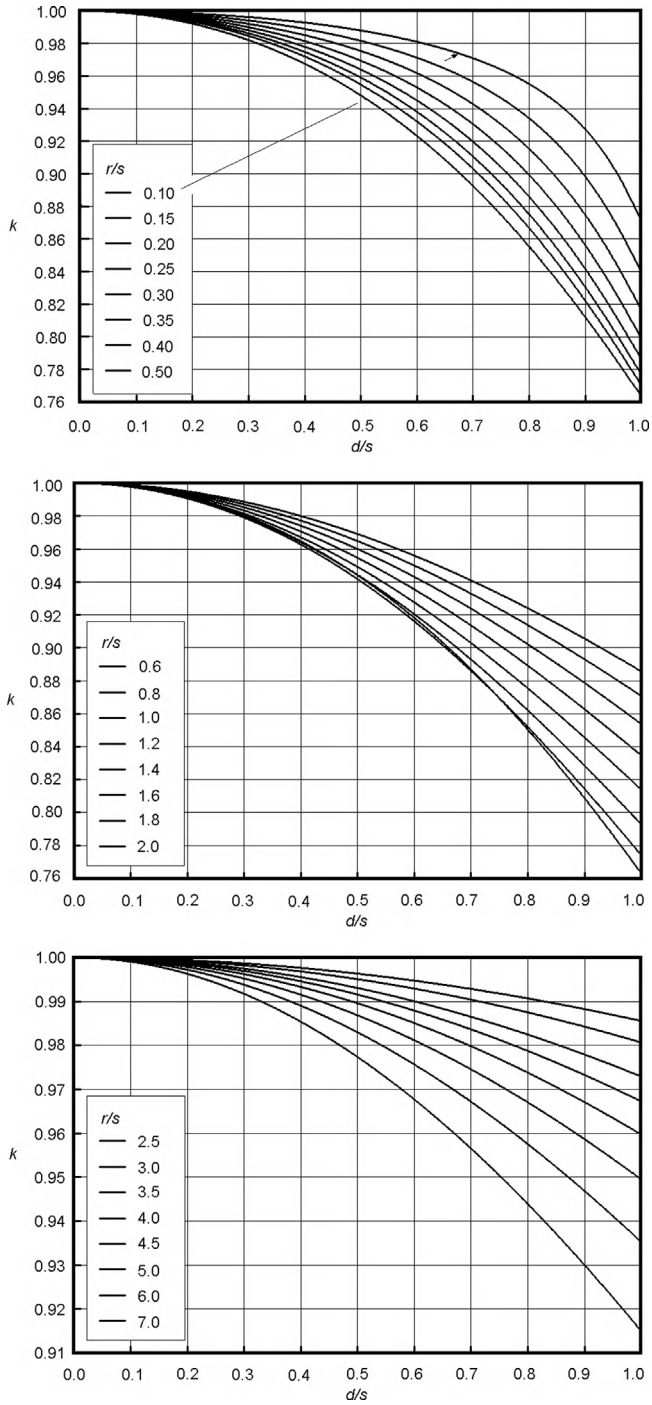


Figure 7.2 Correction factor for finite source radius, eqn. (7.4).

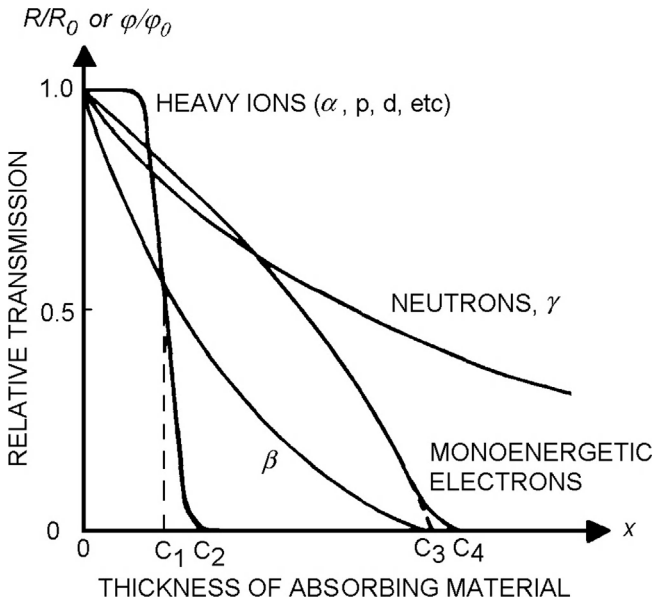


Figure 7.3 Curves showing relative transmission φ/φ_0 (or R/R_0) as function of absorber thickness x. C_1 and C_3 are average, C_2 and C_4 maximum range.

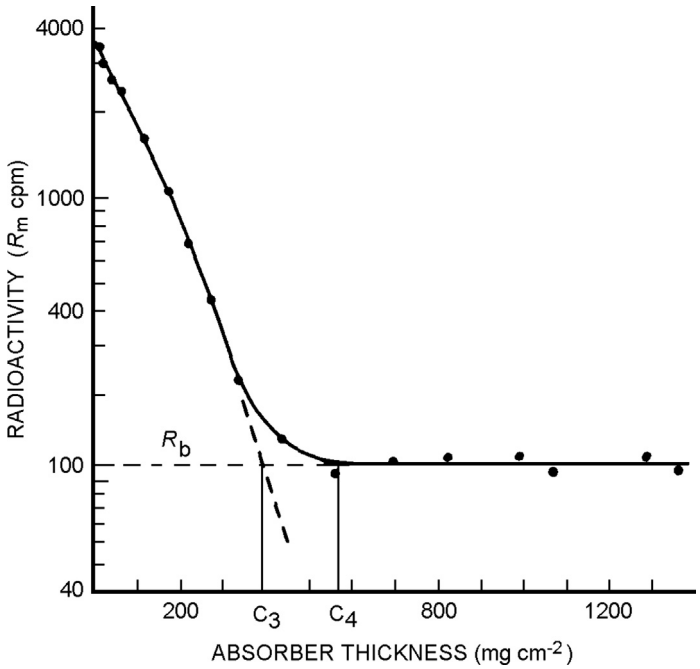


Figure 7.4 Absorption curve for ^{32}P β-radiation showing extrapolated (C_4) and average (C_3) ranges. The dashed curve is obtained after subtraction of background.

background activity R_b, which has to be subtracted from the measured value R_m to obtain the true value for the radiation (e.g. from ^{32}P): $R = R_m - R_b$. The extrapolation of R to a value equal to R_b (i.e. C_3) gives the range.

Whereas it is possible to specify maximum ranges for charged particles, this is not possible for neutral particles such as neutrons and γ-quanta. If the absorber is not too thick, these particles undergo only one collision, or at the most a few, before they are absorbed. As a result the absorption curve has an exponential form.

$$\varphi = \varphi_0 e^{-\mu x} \tag{7.8}$$

where μ is the *total attenuation coefficient*. Thus for n and γ we have

$$\psi_{abs}(x) = e^{-\mu x} \tag{7.9}$$

The reduction in intensity of a beam can occur by two mechanisms. One involves the deflection or scattering of the particles from the direct line of path between the source and the detector and is described by the *scattering coefficient* μ_s. The second mode of reduction is the complete transfer of the projectile energy to the absorbing material (the particles are "captured") and is designated by the (*energy*) *absorption coefficient* μ_a. The (total) attenuation coefficient in Eqn (7.8) is the sum of both these mechanisms.

$$\mu = \mu_s + \mu_a \tag{7.10}$$

Both μ_s and μ_a can be measured independently. The (total) attenuation coefficient is of primary interest in radiation shielding, while the (energy) absorption coefficient is important in considering the effect of radiation on matter.

7.3. ABSORPTION OF PROTONS AND HEAVIER IONS

The mode of interaction of protons and heavier charged particles with the atoms of the absorbing material can be illustrated by considering the absorption of α-particles. With rare exception, α-particles emitted by radioactive nuclides have energies between 4 and 9 MeV. Since the α-particles are so much heavier than electrons, they are deflected very slightly when their Coulomb fields interact with atoms or molecules to form ion pairs. As a result, α-particles travel in a straight line as they pass through matter, which explains the straight paths observed for α-particles in cloud chamber photographs (Fig. 7.5). This is in contrast to the very curved or irregular paths of the secondary electrons emitted in the formation of the ion pair. For a 5 MeV α-particle the maximum energy of the secondary electrons is 2.7 keV. However only a small fraction of the secondary electrons actually receive this much energy; the average energy of the secondary electrons is closer to 100 eV. The ionization caused by more energetic secondary electrons is usually referred to as *δ-tracks* (cf. §8.2).